



SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE
Constituent of Symbiosis International (Deemed University), Pune

Name of Student: Anurag Thakur	PRN:25070122040
--------------------------------	-----------------

Semester: III	Year : AY 2026-27
---------------	-------------------

Subject: Programming with Java

ASSIGNMENT No : 14

TITLE : Exception Handling using Try-Catch
--

Problem Statement

Write a Java program to demonstrate exception handling using try and catch blocks, nested and multiple try-catch blocks.

Learning Objectives

To handle runtime exceptions using try-catch, nested try, and multiple catch blocks.

Theory

Exception handling prevents abnormal termination of programs due to runtime errors. The try block contains code that may generate exceptions, while catch blocks handle specific exceptions. Nested and multiple try-catch structures allow complex error management. Exception handling improves software reliability and user experience by gracefully managing unexpected situations.

Programme

```
import java.util.Scanner;

public class ATM {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        double balance = 10000;

        try {
            System.out.print("Enter withdrawal amount: ");
            double amount = sc.nextDouble();
```

```
        if (amount <= 0) {
            throw new Exception("Withdrawal amount must be greater than
zero.");
        }

        if (amount > balance) {
            throw new Exception("Insufficient balance.");
        }

        balance = balance - amount;

        System.out.println("Withdrawal successful.");
        System.out.println("Remaining balance: " + balance);
    }
    catch (Exception e) {
        System.out.println("Error: " + e.getMessage());
    }

    sc.close();
}
}
```

```
import java.util.Scanner;

public class OnlineShopping {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        try {
            System.out.print("Enter product quantity: ");
            int quantity = sc.nextInt();
        }
```

```

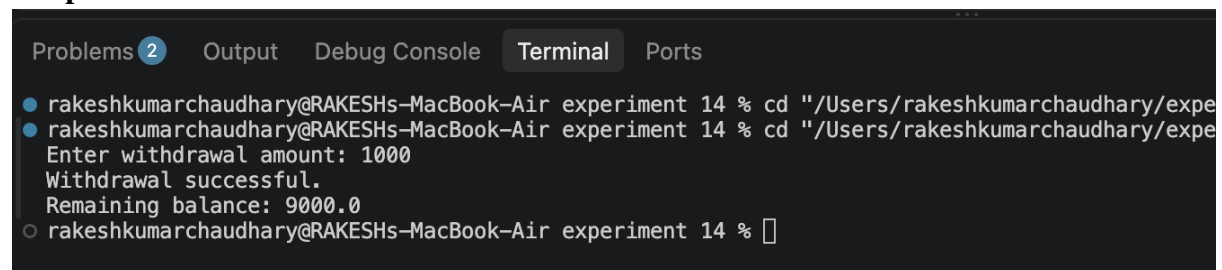
        if (quantity <= 0) {
            throw new Exception("Quantity must be greater than zero.");
        }

        System.out.println("Product quantity: " + quantity);
        System.out.println("Order placed successfully.");
    }
    catch (Exception e) {
        System.out.println("Error: " + e.getMessage());
    }

    sc.close();
}
}

```

Output

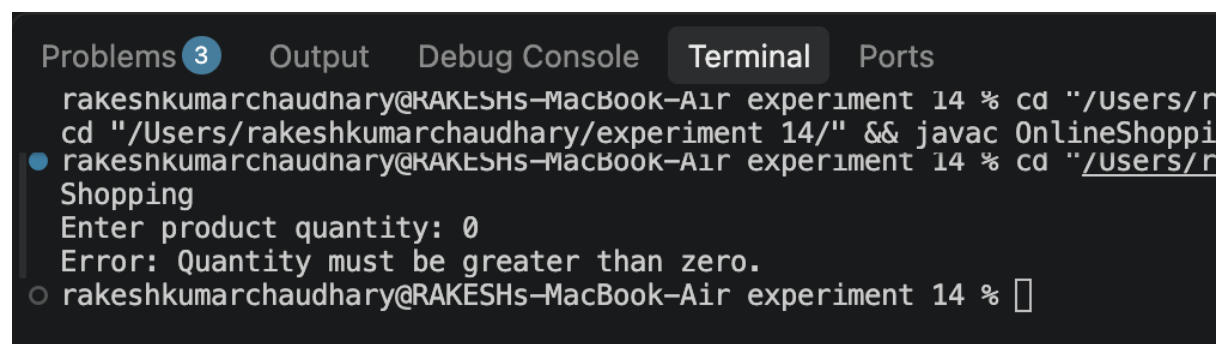


The screenshot shows an IDE terminal window with the 'Terminal' tab selected. It displays the execution of a Java program. The user enters '1000' for the withdrawal amount, and the program outputs 'Withdrawal successful.' and 'Remaining balance: 9000.0'.

```

Problems 2 Output Debug Console Terminal Ports
rakeshkumarchaudhary@RAKESHs-MacBook-Air experiment 14 % cd "/Users/rakeshkumarchaudhary/expe
rakeshkumarchaudhary@RAKESHs-MacBook-Air experiment 14 % cd "/Users/rakeshkumarchaudhary/expe
Enter withdrawal amount: 1000
Withdrawal successful.
Remaining balance: 9000.0
rakeshkumarchaudhary@RAKESHs-MacBook-Air experiment 14 % 

```



The screenshot shows an IDE terminal window with the 'Terminal' tab selected. It displays the execution of a Java program where the user enters '0' for the product quantity. The program catches the exception and outputs 'Error: Quantity must be greater than zero.'.

```

Problems 3 Output Debug Console Terminal Ports
rakeshkumarchaudhary@RAKESHs-MacBook-Air experiment 14 % cd "/Users/r
cd "/Users/rakeshkumarchaudhary/experiment 14/" && javac OnlineShoppi
rakeshkumarchaudhary@RAKESHs-MacBook-Air experiment 14 % cd "/Users/r
Shopping
Enter product quantity: 0
Error: Quantity must be greater than zero.
rakeshkumarchaudhary@RAKESHs-MacBook-Air experiment 14 % 

```

Conclusion

The program successfully demonstrates exception handling using try-catch blocks to handle runtime errors and invalid inputs without abnormal termination.